PL300 Power BI Certification Prep Series:

Visualizing Data and Power BI Service

Lakshmi Ponnurasan

Lakshmi Ponnurasan

Skills Weightage

- Prepare the data (25–30%)
- Model the data (25–30%)
- Visualize and analyze the data (25–30%)
- Manage and secure Power BI (15–20%)

Hi, It’s me!

Santhana Lakshmi Ponnurasan

Power BI DataViz 2025 World Championship Finalist

Microsoft CERTIFIED POWER BI DATA ANALYST ASSOCIATE

MVP Microsoft® Most Valuable Professional

MISSOURI S&T Data Analyst- Institutional Effectiveness

Report Design Requirements

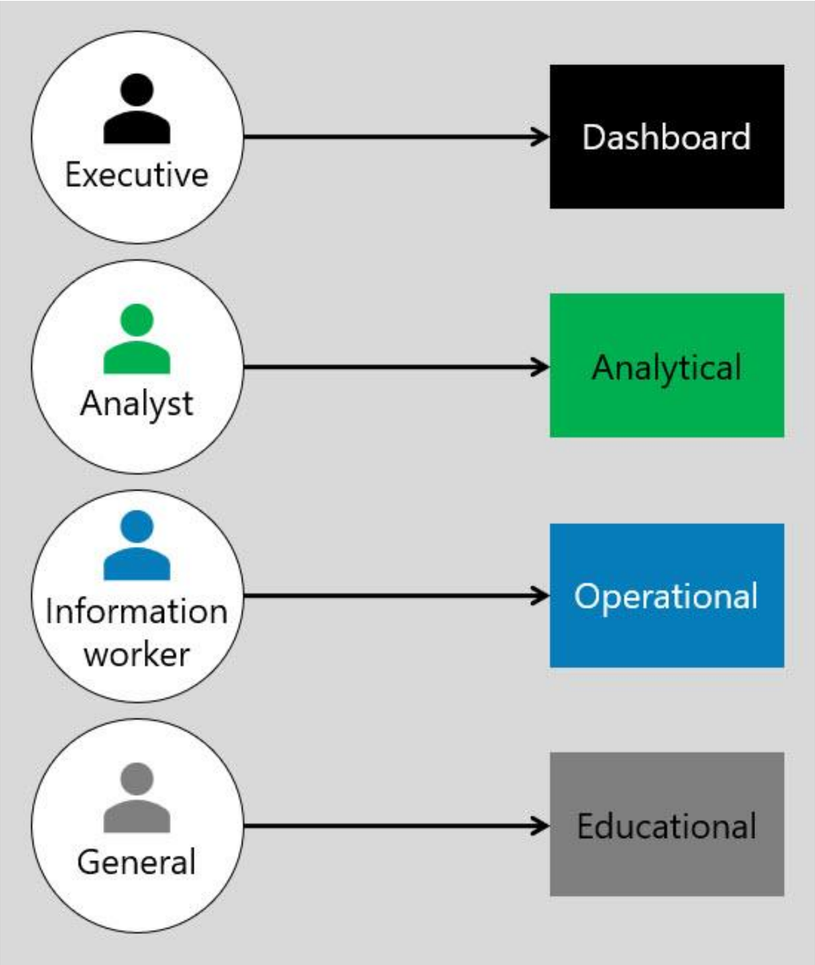
Report Design Requirements

1. Audiences of your reports

There are three broad report consumer audiences:

- **Executive**- responsible for making plans and decisions that typically involve a medium or long-term focus (C-level executives)
- **Analyst**- developing or improving processes, or implementing change. (data analyst)
- **Information worker**-uses data to help make decisions or take actions (inventory manager)

2. Determine Report Types



Report Design Requirements

3. Define UI Requirements

How reports are consumed and to the appearance and behavior of reports.

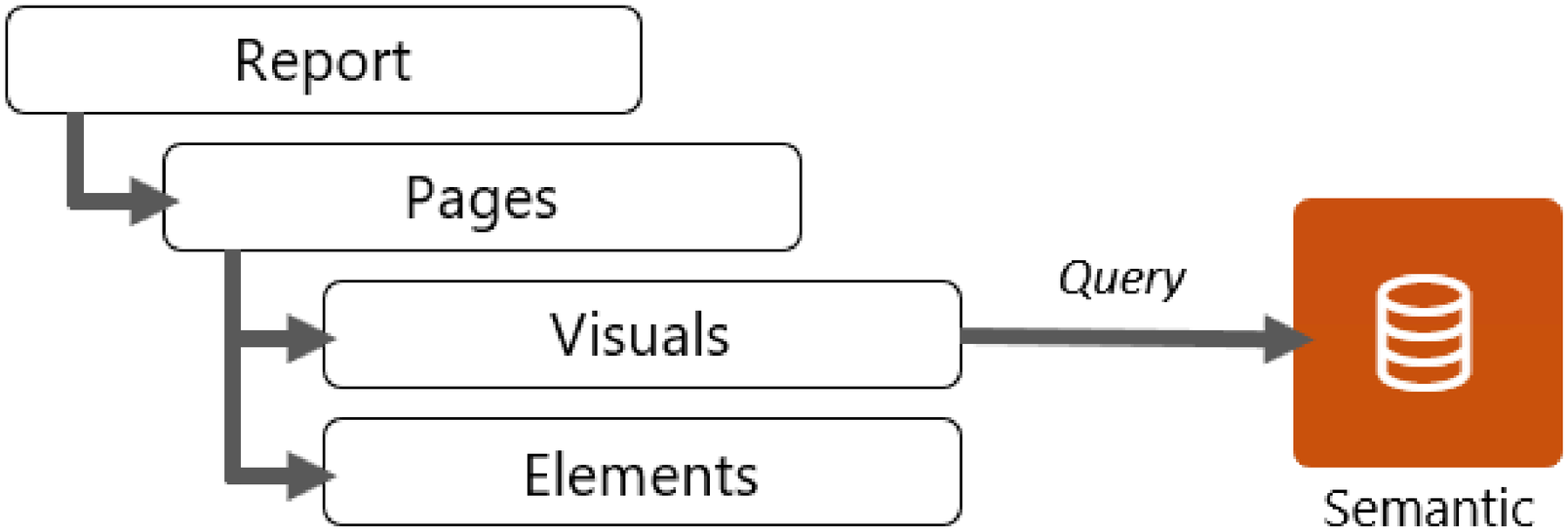
4. Define UX Requirements

Focusing on user experience

Design Power BI Reports

Lakshmi Ponnurasan

Report Structure



Elements include text boxes, buttons, shapes, and images.

Design Analytical Report Layout

Report design is a blend of science and art

1. Placement
2. Alignment
3. Balance
4. Proximity
5. Contrast
6. Repetition

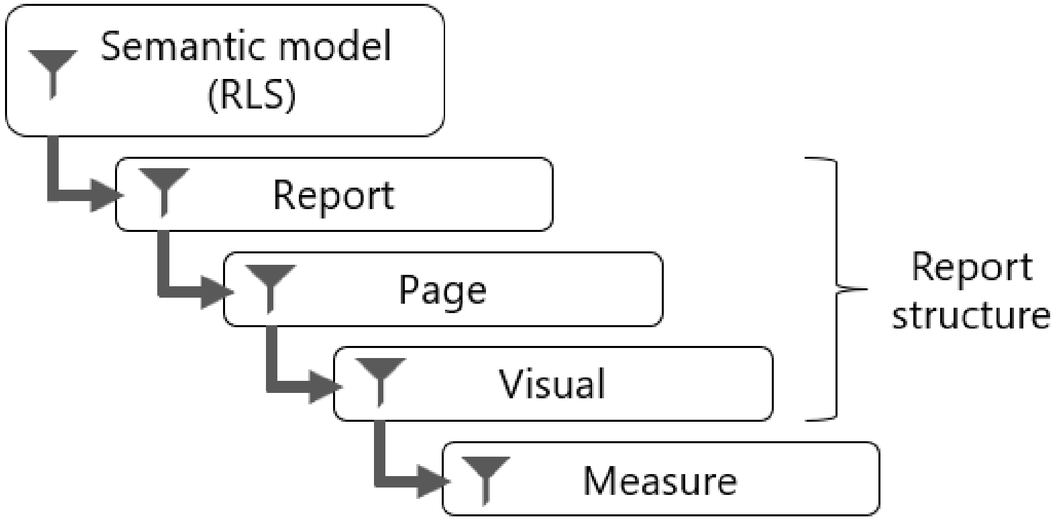
Design Visually Appealing Reports

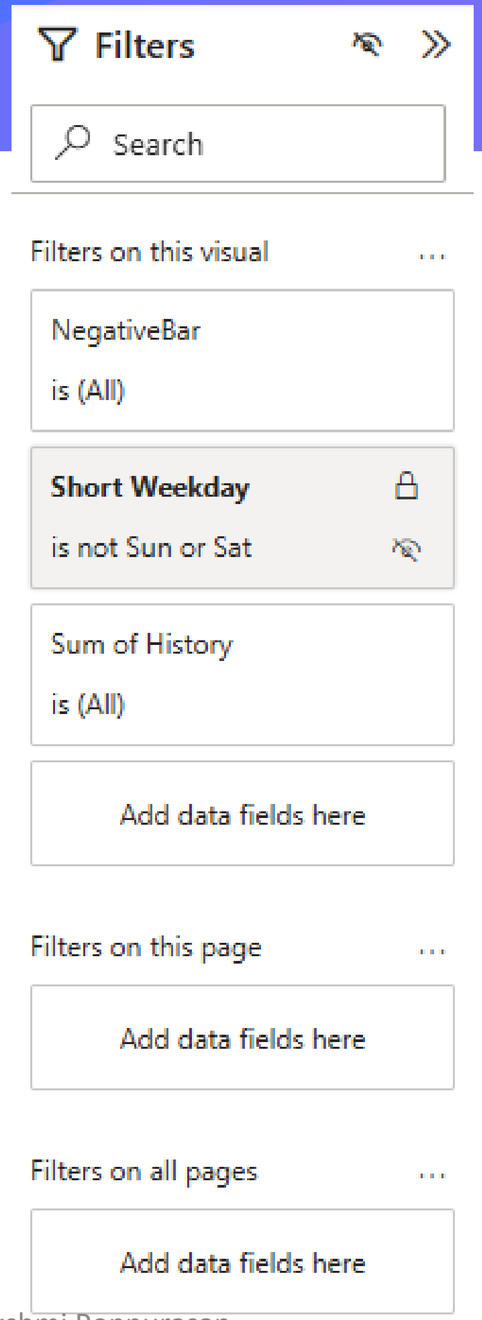
1. Space
2. Margins
3. Size- Canvas and Visual
4. Alignment
5. Colors
6. Consistency

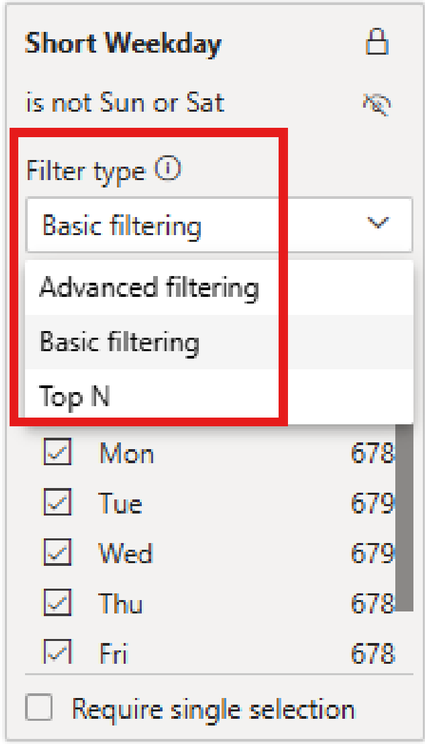
Slicers and Filters

Feature	Slicer	Filter
What it is	A visual placed on the report page for users to interact with	A settings panel option that restricts data shown in visuals
Visibility	Visible to end users	Not always visible (depends on pane settings)
User Interaction	Highly interactive; users click buttons, dropdowns, lists	Limited interaction; mainly for report designers
Where Used	On the **report canvas**	Filter pane (visual-level, page-level, report-level)
Purpose	Let users explore and slice data easily	Control what data appears in visuals or pages
Affects	Visuals on the page (unless synced across pages)	Visual, page, or entire report depending on level
Styling	Fully customizable (fonts, colors, orientation, buttons)	Minimal styling options
Best For	End-user filtering experience	Behind-the-scenes control and refining data context

Filters

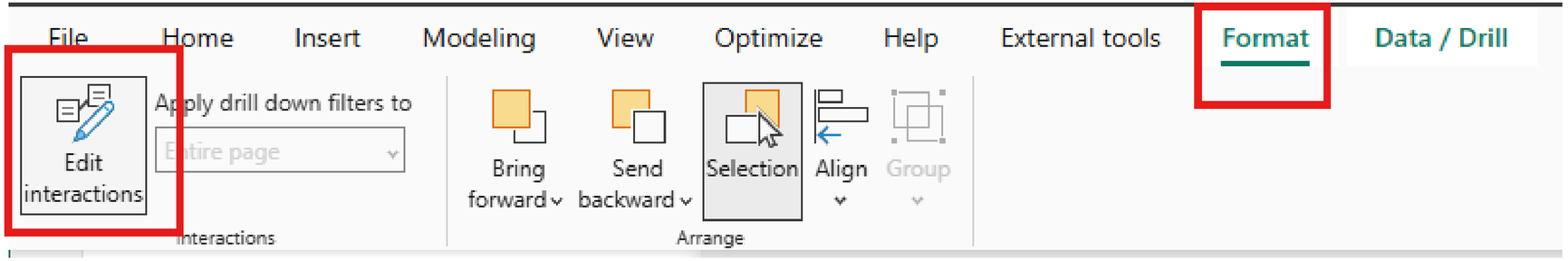

Semantic model (RLS)
Report
Page
Visual
Measure
Report structure


Filters
Search
Filters on this visual
NegativeBar
is (All)
Short Weekday
is not Sun or Sat
Sum of History
is (All)
Add data fields here
Filters on this page
Add data fields here
Filters on all pages
Add data fields here


Short Weekday
is not Sun or Sat
Filter type
Basic filtering
Advanced filtering
Basic filtering
Top N
Mon 678
Tue 679
Wed 679
Thu 678
Fri 678
Require single selection

Slicers *filter other visuals*

Slicer



1. Cross Highlight
2. Cross Filter
3. No interaction

Other Filtering Techniques

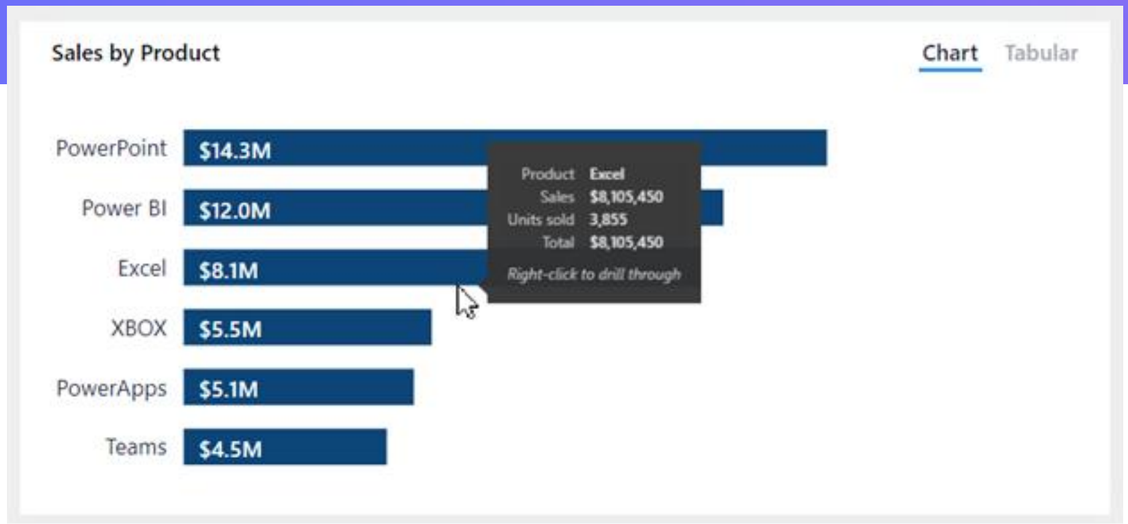
1. Visual Interactions (Cross filter/cross highlight)
2. Drillthrough (drill from visuals)
3. Report Tooltip (Appears on Hover)
4. Bookmarks (capture a specific view of a report)
5. Filter Pane
6. Query Reduction Options

Enhance Power BI UX

Lakshmi Ponnurasan

Basic UX Flow

- High-level metrics
  - Supporting visuals
    - Details, when required



Techniques:

- Use drillable visuals (Drilldown)
- Add tooltips
  - Visual Tooltips
  - Page Tooltips
- Add drillthrough

Manage and Secure Power BI

Lakshmi Ponnurasan

Imagine…

📊 You spent weeks building a detailed sales performance report in Power BI Desktop, filled with **interactive visuals** and **valuable insights**.

→

👬 Now, your leadership team wants to have **access** to this report, and your colleagues from different departments need to **collaborate** on the data.

⚠️ However, sharing a static Power BI (.pbix) file through email isn't a scalable solution:
- creates version control issues
- risks data inconsistency
- makes collaboration difficult.

This is where the Power BI service comes in.

By **publishing** your report to the Power BI service,

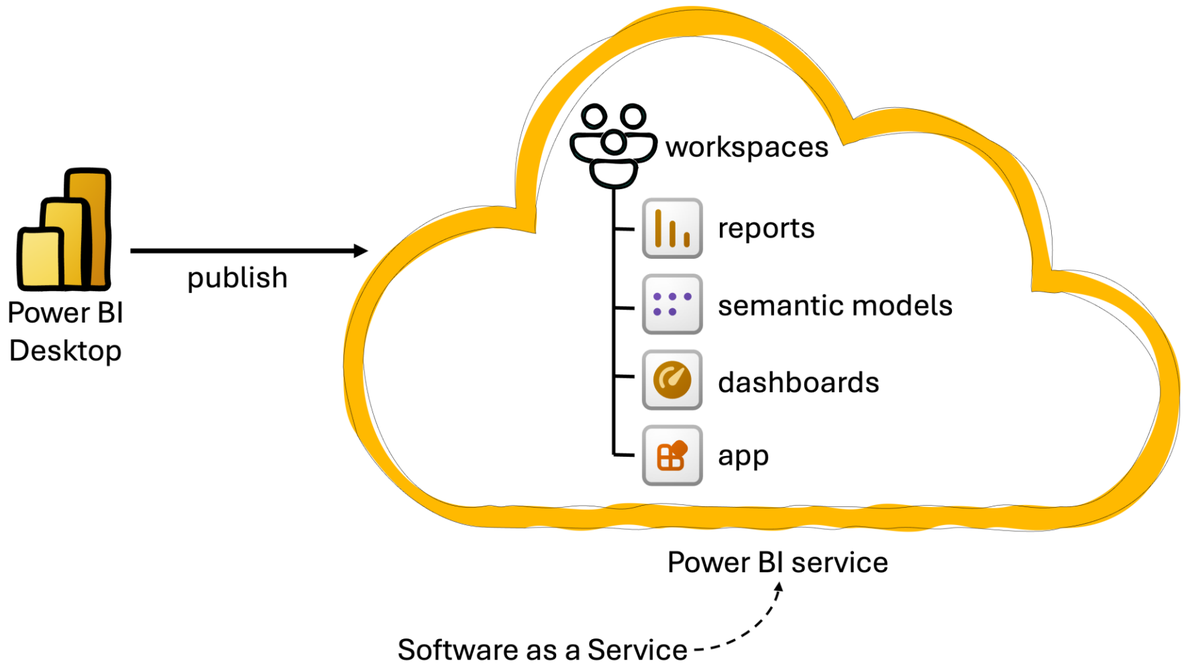
↓

You can create a **centralized**, **web-based** environment

↓

where users can **securely view**, **interact** with, and **collaborate** on reports.

Power BI Service



It's the Software as a Service (SaaS) part of Power BI and can be accessed through a web browser. https://app.powerbi.com

Power BI workflow

Building a report in **Power BI Desktop**

Publishing it to the **Power BI service**

Building blocks of Power BI Service:

- **W**orkspaces
- **A**pps
- **R**eports
- **D**ashboards
- **S**emantic models

W.A.R.D.S

Workspaces

Workspaces are like **folders/container** where all your Power BI content lives.

They help you **organize** and **manage** things like reports, semantic models, and dashboards in **one place**.

There are two types of workspaces:
1. My Workspace, which is private to an individual user

2. Shared workspaces, which can be accessed by multiple users with assigned roles.

Reports

A report has **one or more pages**, each **containing one or more visuals.**

Presenting data as visuals makes it easier to grasp the meaning of what you're seeing.

And, the **visuals aren't static**. You interact with the visuals and filters as you dig into the data to discover insights and find answers.

Semantic Models

Power BI semantic models represent a **source of data** that's ready for reporting and visualization.

The semantic model is a **collection of tables, relationships, and measures** that define how data is structured and how it can be used in reports.

Semantic models can be **promoted** or even **certified** to help users find the right data for their reports.

Dashboards

- Dashboards are **single-page** visual summaries that can **pull data from multiple reports and models.**
- Focus on **quick insights**.
- Support **alerts** on certain tiles.
- Dashboards are exclusive to the **Power BI service.**

Create a dashboard:
By pinning a visual from a report, which is called a ***tile***.

A dashboard can contain tiles from multiple reports, and it can be shared with other users in the Power BI service.

Dashboards vs Reports

Feature	Dashboards	Reports
Pages	Single page	Multiple pages
Data Sources	Can pull from **multiple reports/models**	Uses **one** semantic model
Interactivity	Limited (no drill-down)	Highly interactive (filters, drill-down, drill-through)
Alerts	✔ Yes (on certain tiles)	✘ No
Subscriptions & Favorites	✔ Yes	✔ Yes
Natural Language (Q&A)	✔ Yes	✔ Yes
Where Available	Only in **Power BI Service**	Power BI Desktop & Service
How Created	By **pinning visuals** as tiles	Built directly in Desktop
Best Use	Quick highlights & at-a-glance insights	Detailed analysis & deep exploration
Content Mix	Can include tiles from **multiple reports**	All visuals come from the **same model/report**

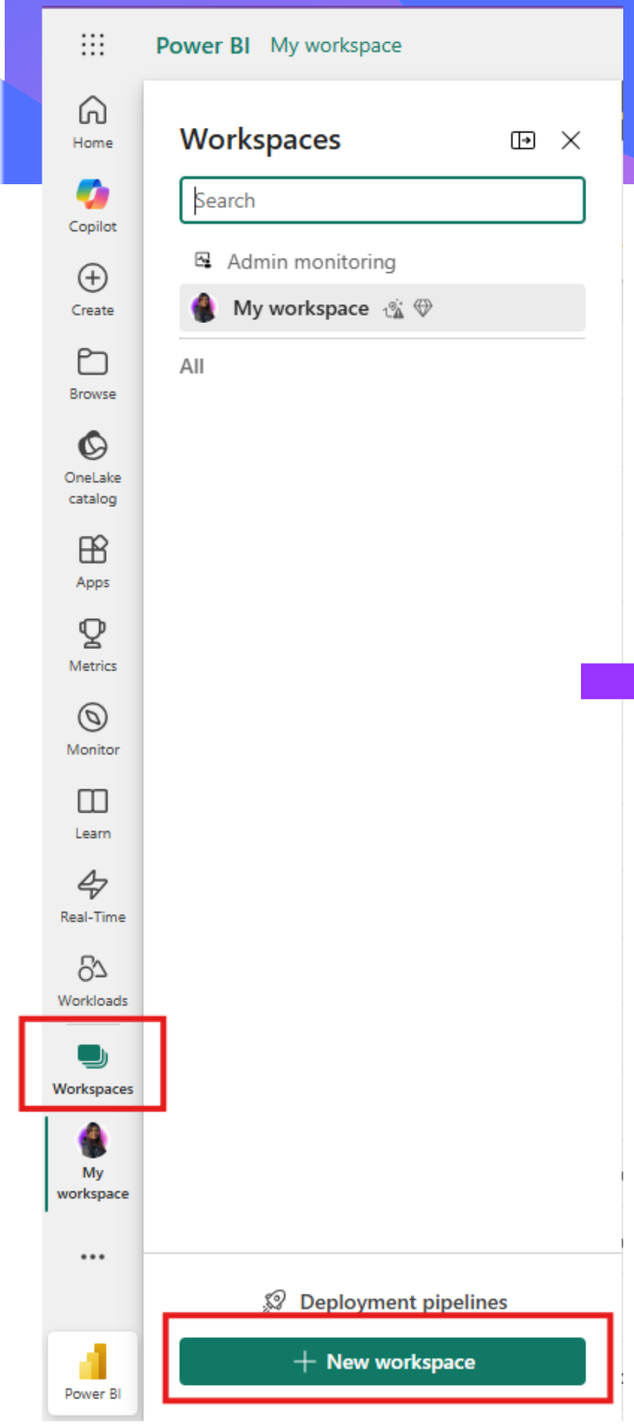
Apps

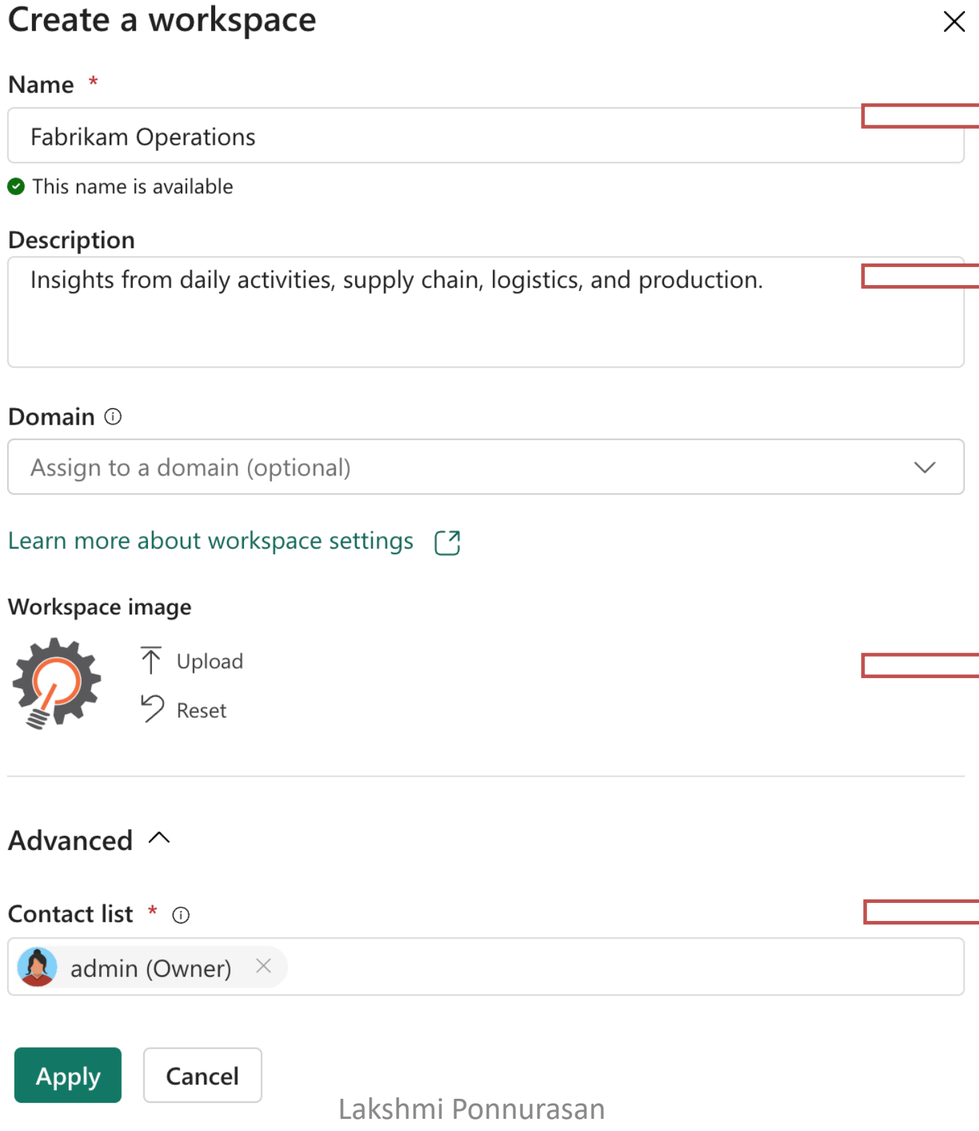
- Apps are an easy way for designers to **share related content**, like dashboards, semantic models, and reports, **all in one place**.

- Apps are a great way to share content with a **larger audience**, such as an entire department or organization. You can even target multiple audiences in the same app.

Workspaces

Create a workspace





Choose a **unique** name for it

Optional description to help others understand the purpose of the workspace.

Can be uploaded as a **.png or .jpg file.** File size must be **less than 45 KB.**

The names of people to contact for information about the workspace. By default, the workspace admins are the contacts.

Lakshmi Ponnurasan

License Modes

The license mode determines **how** the content in the workspace can be shared and consumed.

License mode

- (●) **Pro**
  Select Pro to use basic Power BI features and collaborate on reports, dashboards, and scorecards. To access a Pro workspace, users need Pro per-user licenses. Learn more
- ( ) **Trial**
  Select the free trial per-user license to try all the new features and experiences in Microsoft Fabric for 60 days. A Microsoft Fabric trial license allows users to create Microsoft Fabric items and collaborate with others in a Microsoft Fabric trial capacity. Explore new capabilities in Power BI, Data Factory, Data Engineering, and Real-Time Intelligence, among others. Learn more
- ( ) **Premium per-user**
  Select Premium per-user to collaborate using Power BI Premium features, including paginated reports, dataflows, and datamarts. To collaborate and share content in a Premium per-user workspace, users need Premium per-user licenses. Learn more
- ( ) **Premium capacity**
  Select premium capacity if the workspace will be hosted in a premium capacity. When you share, collaborate on, and distribute Power BI and Microsoft Fabric content, users in the viewer role can access this content without needing a Pro or Premium per-user license. Learn more
- ( ) **Embedded**
  Select embedded if the workspace will be hosted in an Azure embedded capacity. ISVs and developers use Power BI Embedded to embed visuals and analytics in their applications. Learn more
- ( ) **Fabric capacity**
  Select Fabric capacity if the workspace will be hosted in a Microsoft Fabric capacity. With Fabric capacities, users can create Microsoft Fabric items and collaborate with others using Fabric features and experiences. Explore new capabilities in Power BI, Data Factory, Data Engineering, and Real-Time Intelligence, among others. Learn more

- Pro
- Premium per-user (PPU)
- Fabric

License Modes

License Mode	Who Can Access?	Key Features	Best For	Disadvantage
Power BI Pro	Any user **must have Pro** to access shared content (unless content is placed in Premium capacity).	Create & share reports, collaborate in workspaces, publish apps, scheduled refreshes (basic).	Small / medium teams that all have Pro licenses.	Requires a Pro license for every user who views content; limited model size, fewer refreshes, no Premium/large-model AI features.
Premium Per User (PPU)	Only users with **PPU** can access PPU workspaces and content.	All Pro features **plus** larger model sizes, more scheduled refreshes, some Premium AI features, paginated reports, and advanced performance.	Individuals or small teams needing Premium features without buying capacity.	Every consumer needs a PPU license (cost per user); not as flexible for broad organization-wide sharing; can be costly at scale.
Premium / Fabric Capacity	Content in a Premium/Fabric capacity can be **viewed by free users** in the org (creators still need Pro or PPU depending on the setup).	Dedicated capacity (better performance & scale), very large models, more refreshes, advanced AI, dataflows, pipelines, warehousing, real-time streaming.	Large teams or enterprises that need scale, broad sharing, and enterprise features.	Highest cost (capacity-based); needs capacity administration and planning; more complex to manage.

Once a workspace has been created, Identifies Fabric workspaces and Identifies PPU workspaces

Roles and Access Permissions

Roles let you **manage who can do what** in a workspace, so teams can collaborate.

Assign roles to:

- Individuals
- User groups such as security groups, Microsoft 365 groups, and distribution lists.

4 Role Types:

- Viewers
- Contributors
- Members
- Admins

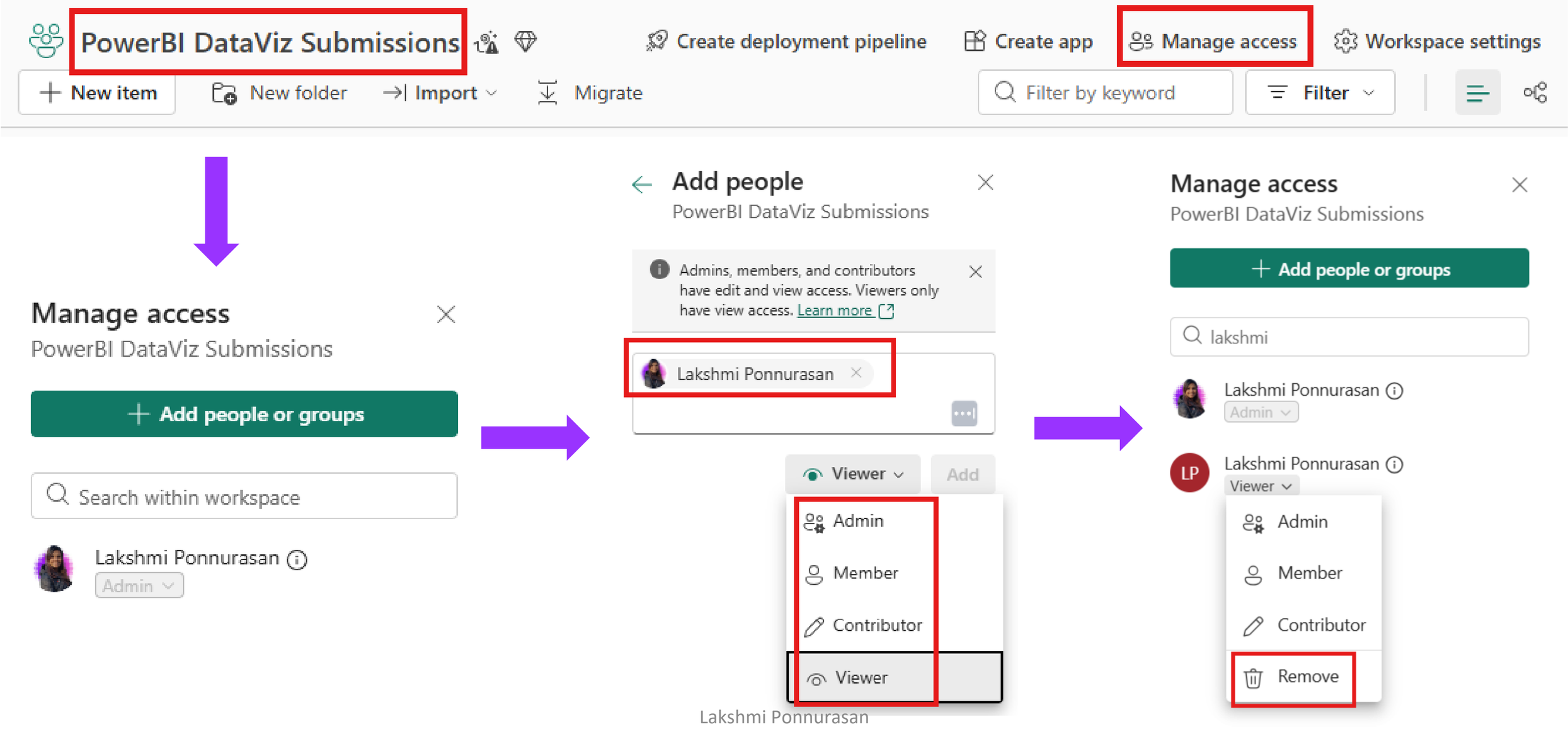
Roles and Access Permissions

Role	What They Can Do	What They *Can't* Do
Viewer	View and interact with reports & dashboards ⭐**This is also the only role where Row-Level Security (RLS) will be enforced**	Create, edit, or delete content
Contributor	Create, edit, delete content	Publish/update apps, change workspace settings
Member	Manage content, publish/update apps, share items, change data settings	Delete workspace, manage Admins
Admin	Full control: manage settings, users, app publishing, delete workspace	— (no restrictions)

Roles and Access Permissions

Capability	Admin	Member	Contributor	Viewer
Update and delete the workspace.	X			
Add/remove people, including other admins.	X			
Add members or others with lower permissions.	X	X		
Publish and update an app.	X	X		
Share an item or share an app.	X	X		
Allow others to reshare items.	X	X		
Create, edit, and delete content in the workspace.	X	X	X	
Publish reports to the workspace, delete content.	X	X	X	
Create a report in another workspace based on a dataset in this workspace.	X	X	X	X [1]
Copy a report.	X	X	X	X [1]
View and interact with an item.	X	X	X	X

Roles and Access Permissions



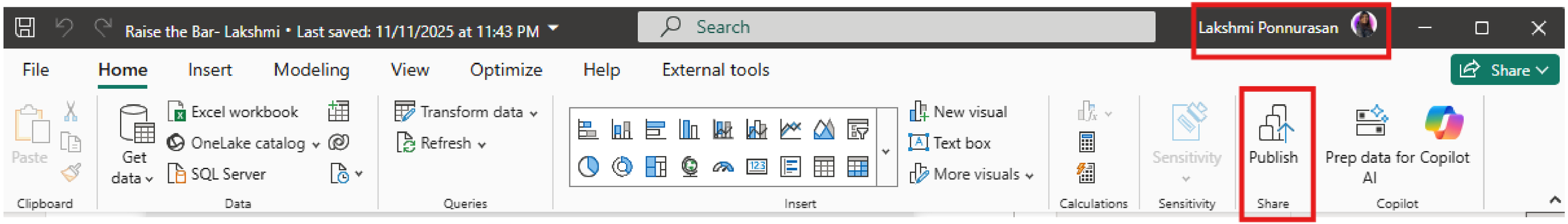
Lakshmi Ponnurasan

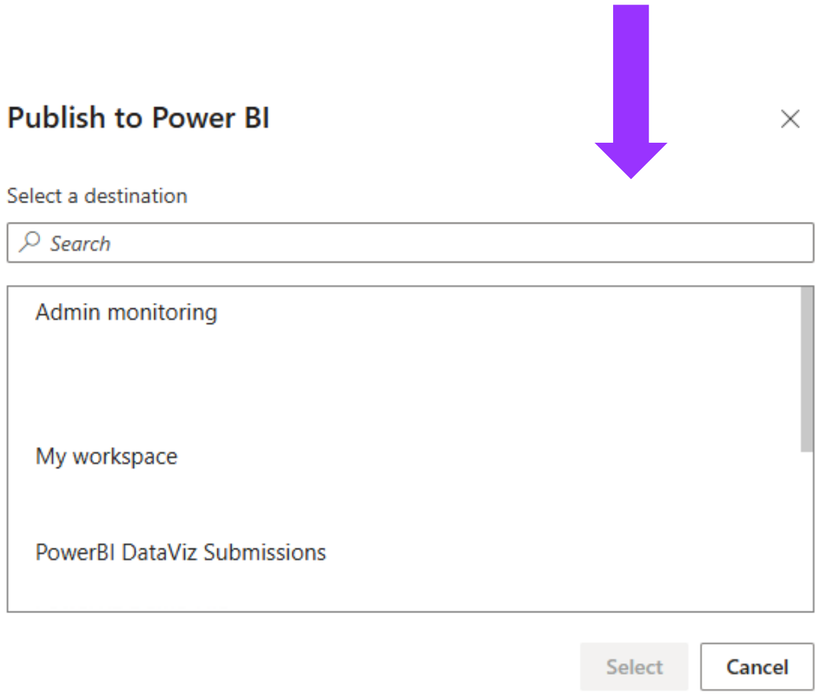
Publishing to Power BI Service

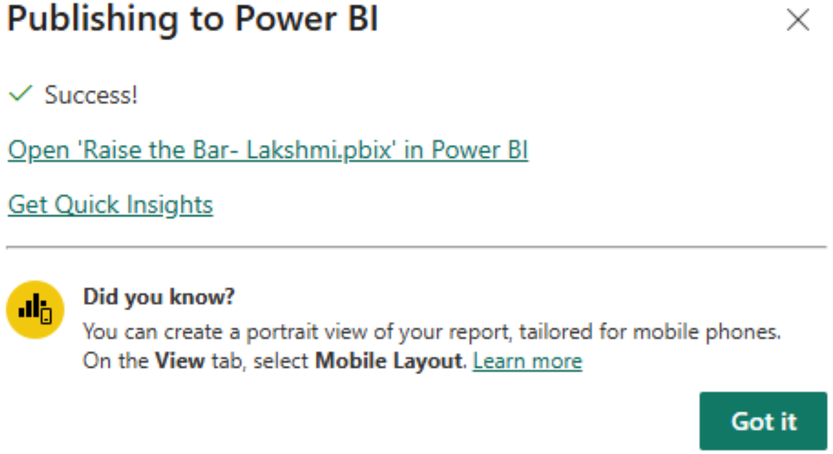
1. Publish from Power BI Desktop
2. Upload a Power BI report (.pbix file) directly to the workspace.

Publishing to Power BI Service

1. Publish from Power BI Desktop





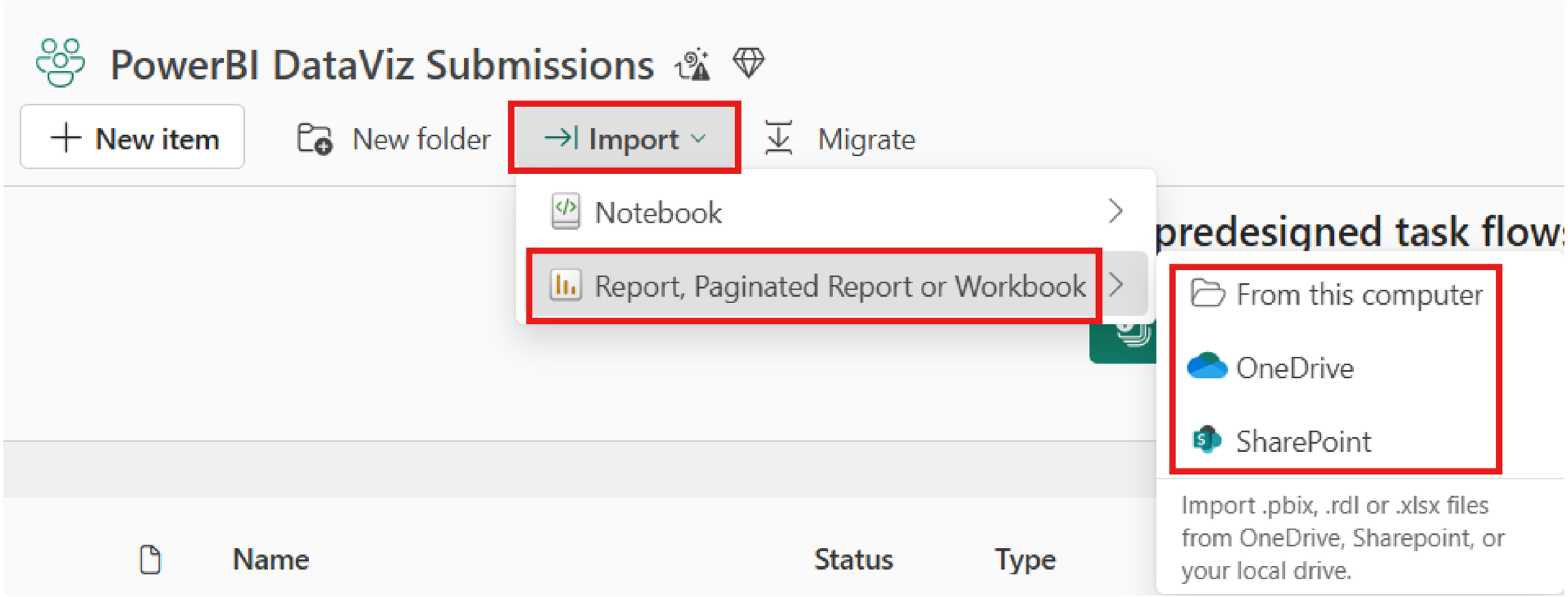


The semantic model is also published with the report. This means that the data model, including tables, relationships, and measures, is now available in the Power BI service.

Lakshmi Ponnurasan

Publishing to Power BI Service

2. Upload a Power BI report (.pbix file) directly to the workspace.



💡 QUIZ

Which one of the following options is the best description of a **workspace**?

1. A workspace is a feature in Power BI service that allows users to view reports only.

2. A workspace is a centralized location or repository that allows you to collaborate with colleagues and teams to create collections of reports, dashboards, and so on.

3. A workspace is a feature that allows you to view and edit the semantic model, build visualizations, and transform the data.

💡 QUIZ

Which of the following statements about **Power BI workspaces** is correct?

1. A Power BI Pro license allows users to create and share workspaces, and anyone with a free Power BI license can access them.

2. Contributors in a workspace can create, edit, and delete content, but they can't manage app publishing or workspace settings.

3. Members have the highest level of control in a Power BI workspace, including managing roles and deleting the workspace.

💡 QUIZ

Which of the following statements about **Power BI Apps** is correct?

1. Power BI Apps are only accessible to users who are members of the workspace where the app was created.

2. Once an app is installed, users must manually search for and open each dashboard or report separately.

3. Apps in Power BI allow designers to share multiple related content items together, reducing the need to share them individually.

Understanding Sharing Models

Lakshmi Ponnurasan

Understanding Sharing Models

1. Workspace Roles:

Viewer
Contributor
Member
Admin

2. Item Level Sharing:

Allows you to **share specific** reports or dashboards with **selected individuals or groups.**

Recipients **can view** and interact with the content but **cannot edit** it.

You can share via **links** or **directly** grant access to specific individuals or groups.

It overcomes the "all-or-nothing" limitation of workspace roles. For example, you can share a single report with external partners without exposing other workspace content. However, it's important to manage permissions carefully to ensure that only authorized users can access the shared content.

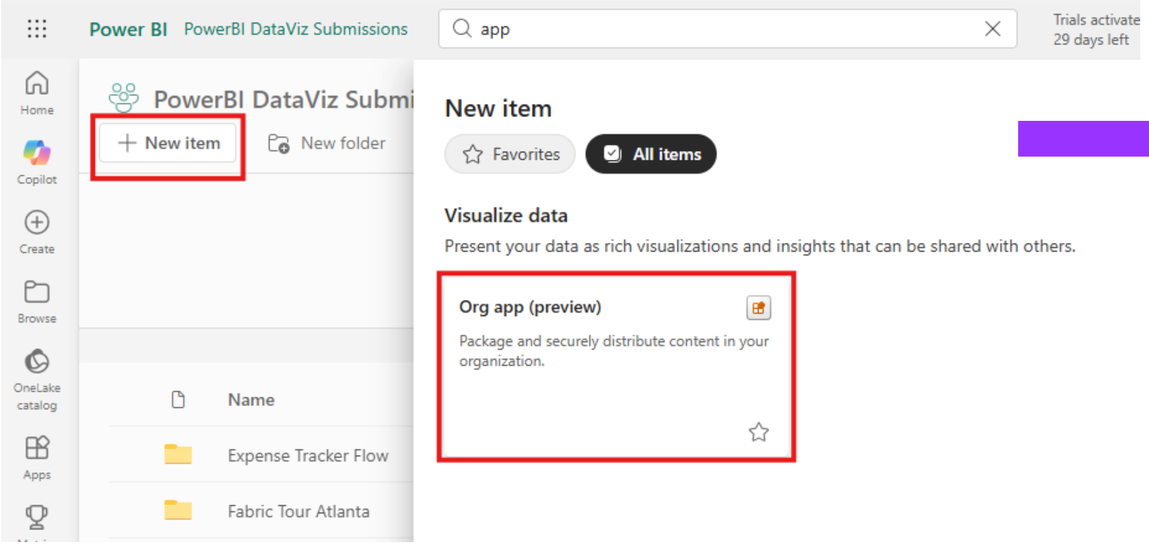
3. Power BI Apps:

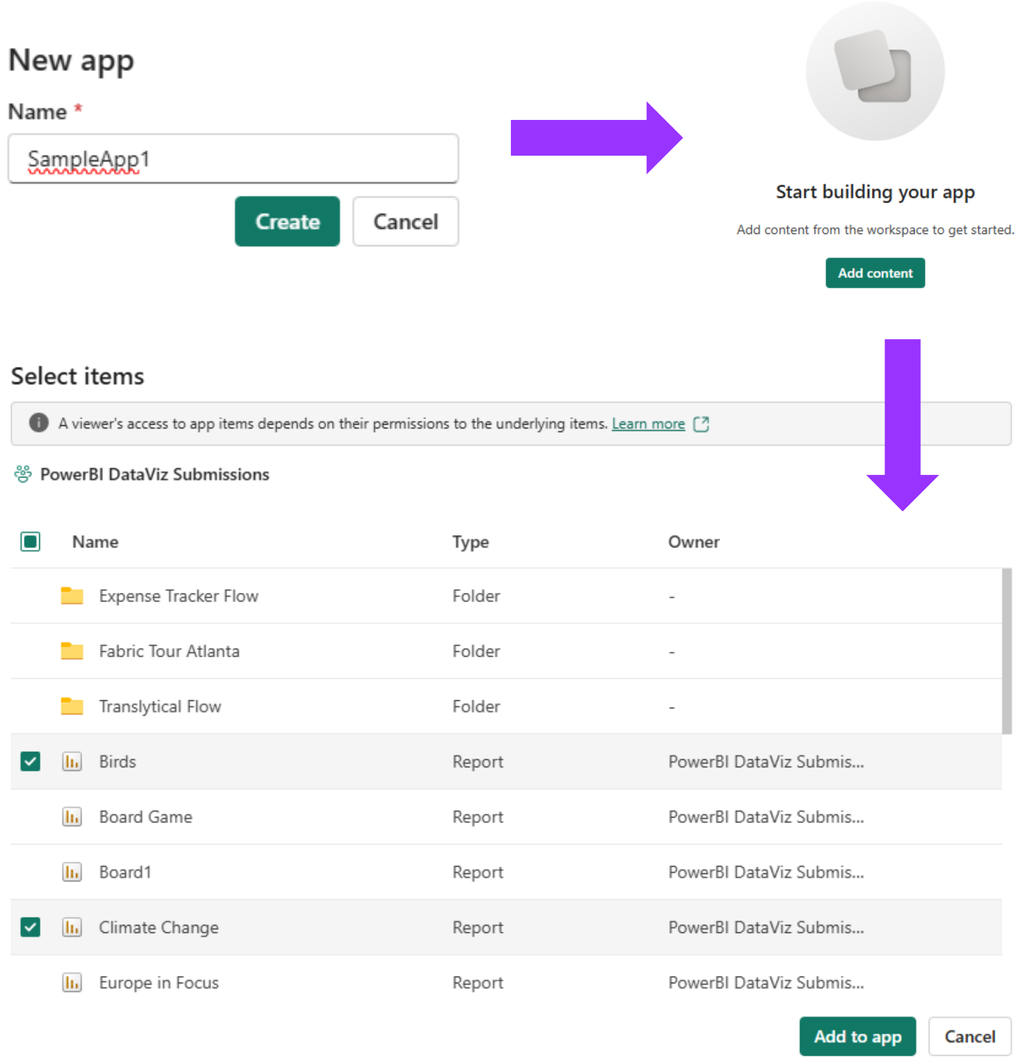
Share collections of dashboards, reports, and other content with a **larger audience without requiring them to be part of your workspace** or sharing individual items separately.

Advantage: **Control it provides when updating content.** The workspace acts as a staging area where changes can be made and tested without immediately affecting the published app. Once the updates are ready, the app can be republished, ensuring that users only see the finalized version.

Note: It's important to note that **each workspace can only have one app**, and the **content included in the app must originate from that workspace**.

Create a Power BI App





Apply Data Governance Principles

1. Endorse your content

- Helps users identify high-quality data.

Types:

1. **Promotion** allows content owners or members with write permissions in a workspace to **highlight content** they believe is valuable and ready for broader use. Promoted content is marked with a **badge**, making it easily recognizable in lists and searches.

2. **Certification** signifies that the **content meets the organization's quality standards** and is deemed authoritative. Only **authorized reviewers**, as defined by the Power BI administrator, can certify content. Certified content is also labeled with a **distinct badge** and is prioritized in searches.

2. Apply a Sensitivity Label

- Sensitivity labels allow users to **tag content with labels** such as "Confidential" or "Highly Confidential”

- Once applied, the label is visible in the workspace and persists when the **content is exported** to supported formats like Excel, PowerPoint, or PDF.

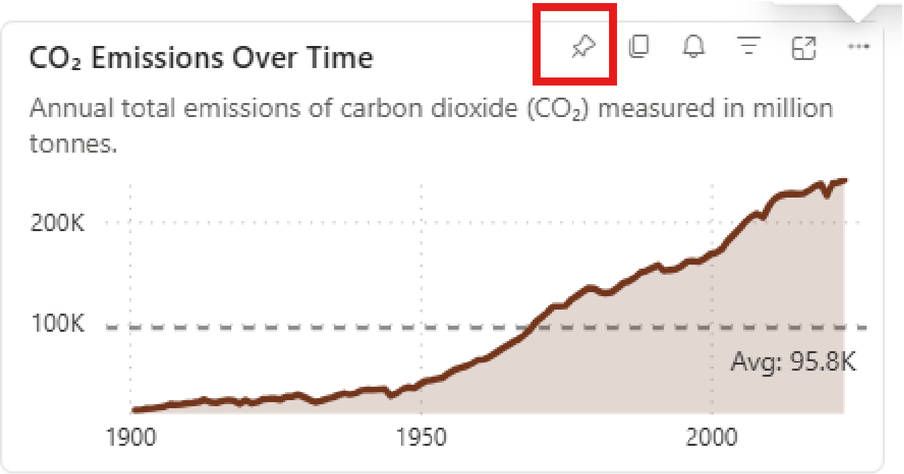
⭐ Monitoring and tracking **usage metrics** and performance in Power BI is essential for understanding how your reports and dashboards are being utilized across your organization.

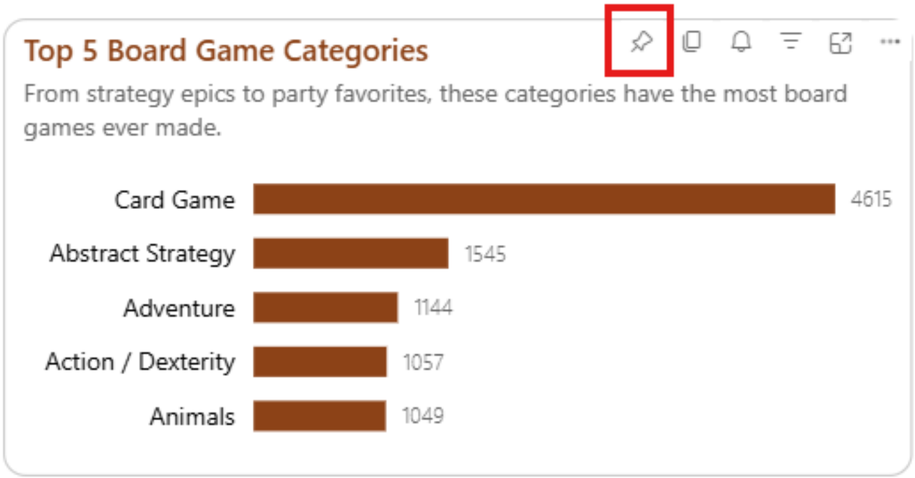
Dashboards in Power BI

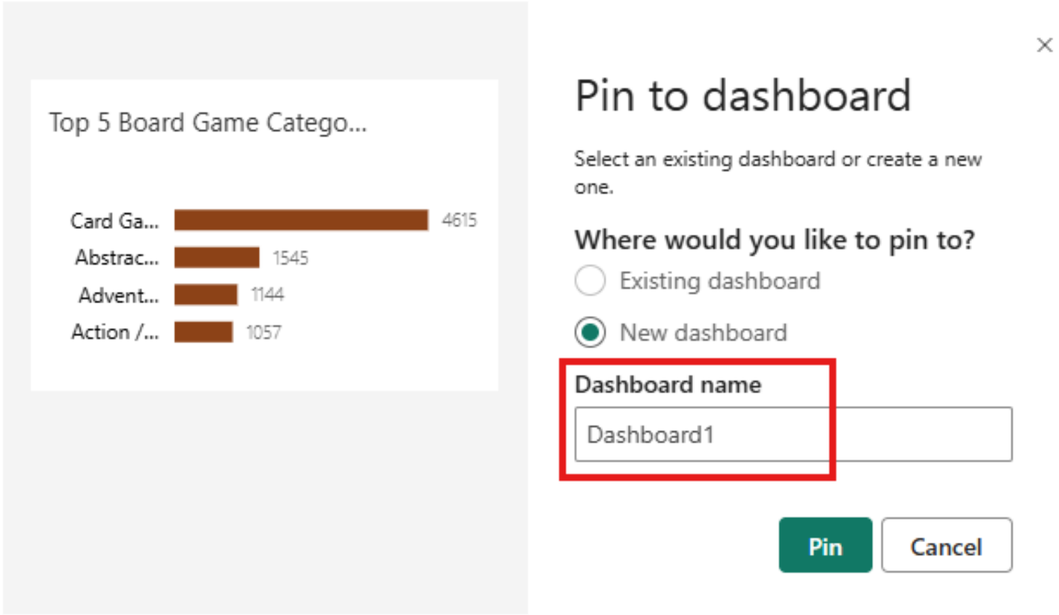
Lakshmi Ponnurasan

Dashboards

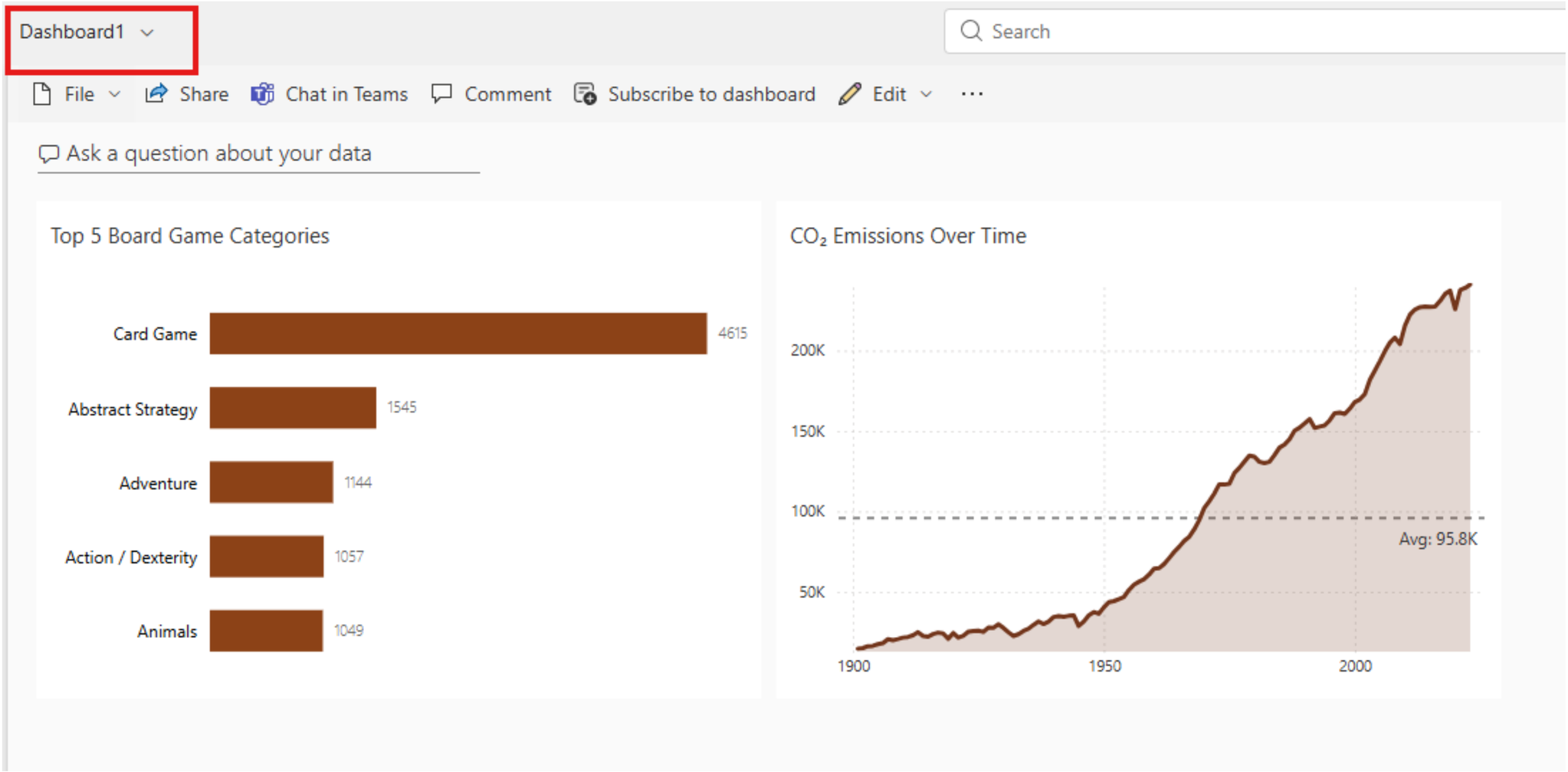
Dashboards are created by ***pinning*** report visuals, and are then called ***tiles*** within reports.





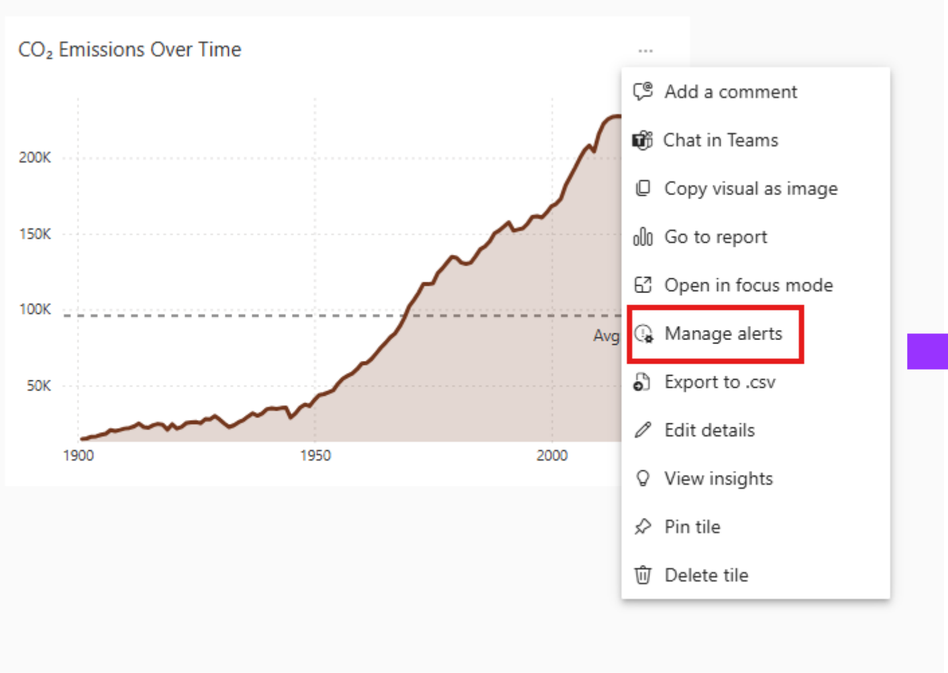


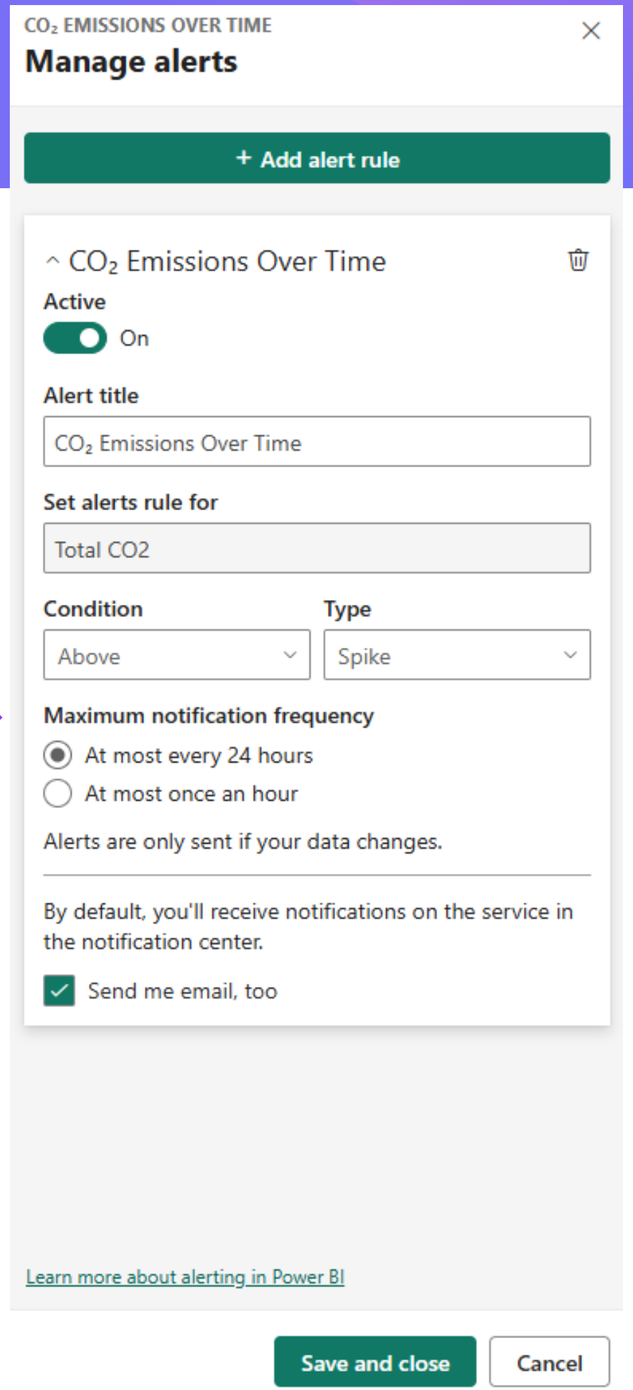
Dashboards


Dashboard1
Search
File
Share
Chat in Teams
Comment
Subscribe to dashboard
Edit
Ask a question about your data
Top 5 Board Game Categories
Card Game
4615
Abstract Strategy
1545
Adventure
1144
Action / Dexterity
1057
Animals
1049
CO_2 Emissions Over Time
200K
150K
100K
50K
Avg: 95.8K
1900
1950
2000

Data Alerts

- Data alerts can **notify** you or a user that a specific data point is above, below, or at a specific threshold that you can set.
- Only available on Power BI service
- They're available on such report elements such as KPI visuals, gauges, and cards.





Secure Data Access in Power BI (RLS,OLS)

Lakshmi Ponnurasan

Imagine…

You work for Tailspin Traders, a company that maintains a database of sales, products, and employee information.

You want to create a single report where employees in a specific department can only view the sales data relevant to their department.

Row Level Security (RLS)

- RLS allows you to **control access to data at the row level**, ensuring that users only see data pertinent to their role.
- As a data modeler, you set up RLS by **creating one or more roles**. A role has a unique name in the model, and it usually includes one or more rules.
- Rules enforce filters on model tables by **using Data Analysis Expressions (DAX) filter expressions.**

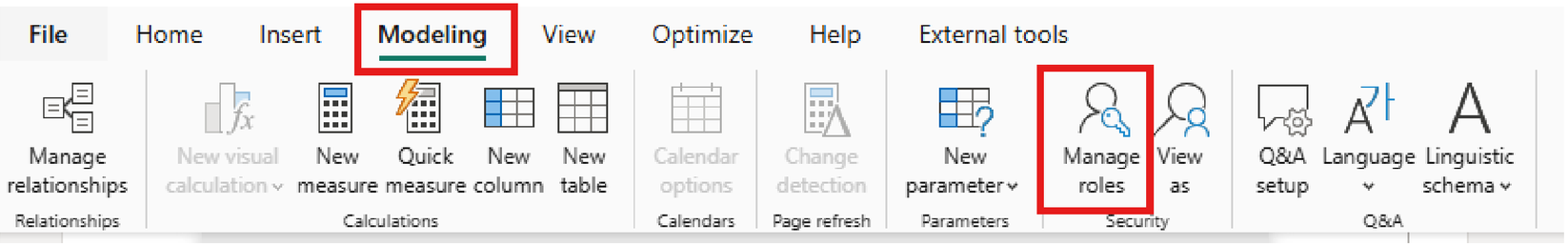
Types

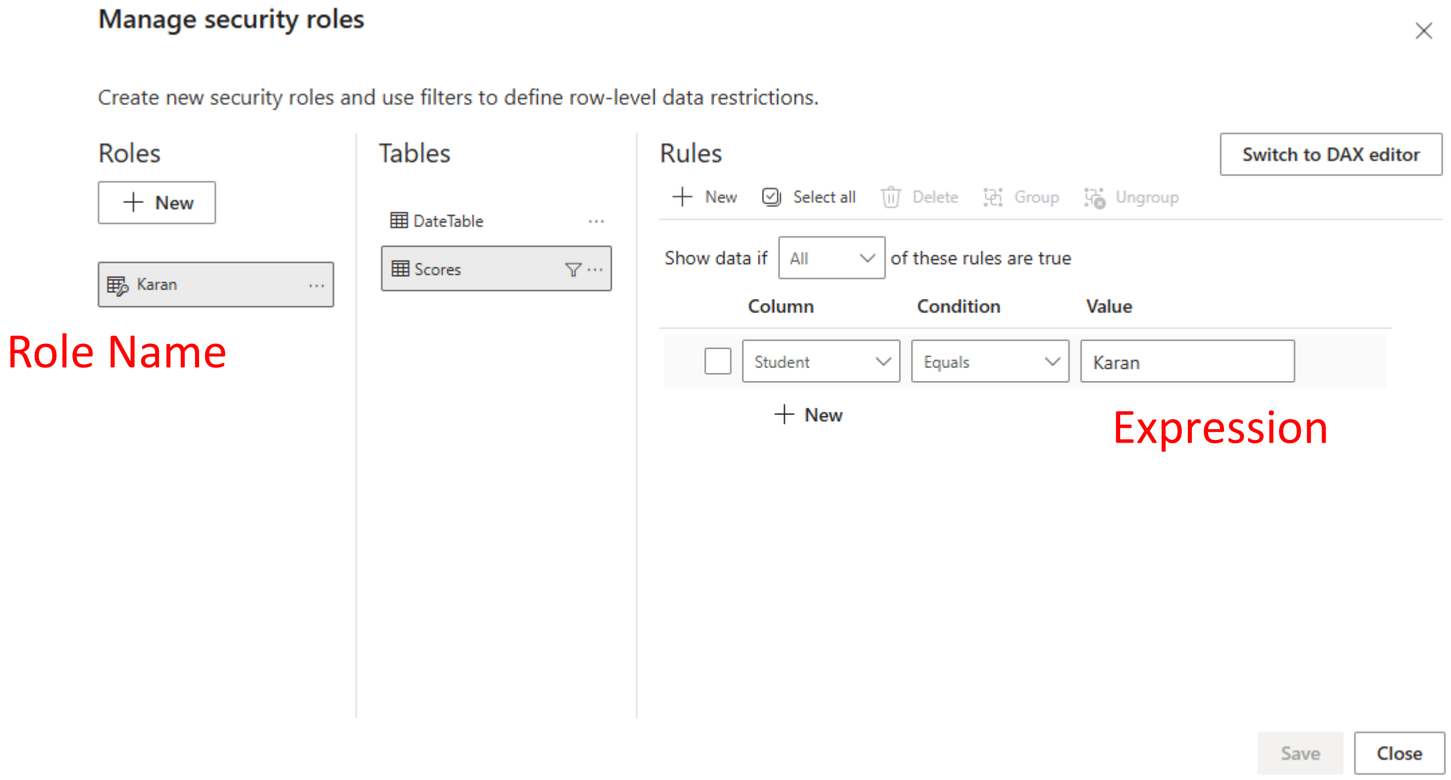
- The **Static** method is suitable for a small number of roles that rarely change.
- The **Dynamic** method is better for a large number of roles that change frequently.
- You can use **single sign-on (SSO)** for some data sources that use Direct Query, enforcing data permissions at the data source level.

⭐ RLS Configuration Steps

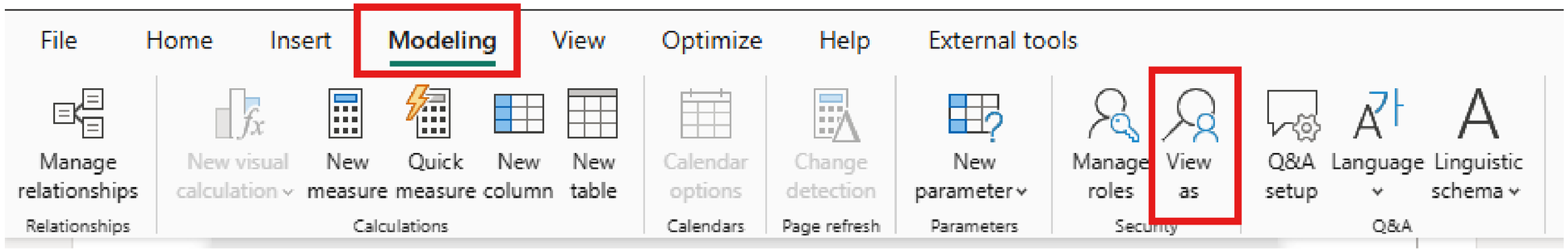
1. Begin by creating a report in Microsoft Power BI Desktop
2. Then, set up RLS roles using DAX
3. Test them within Power BI Desktop.
4. Deploy the report to Microsoft Power BI service
5. Add members to the roles
6. Conduct final testing in the Power BI service.

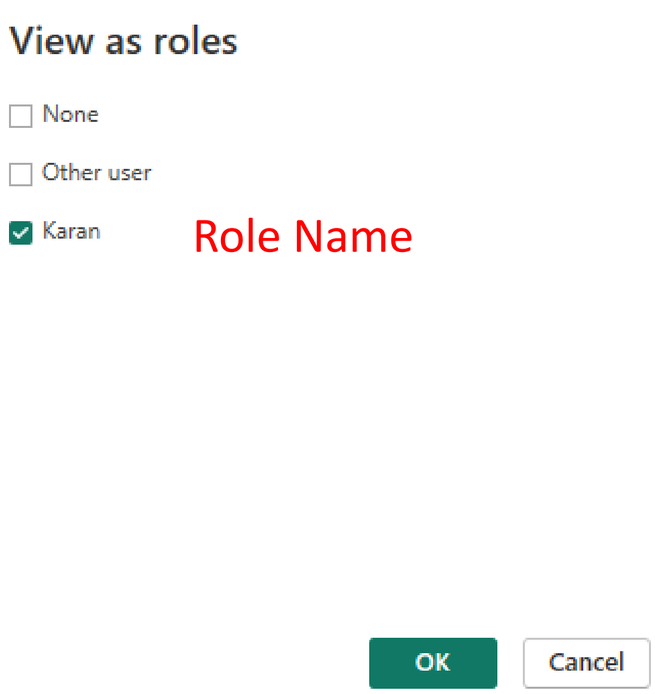
Set up RLS in Power BI Desktop



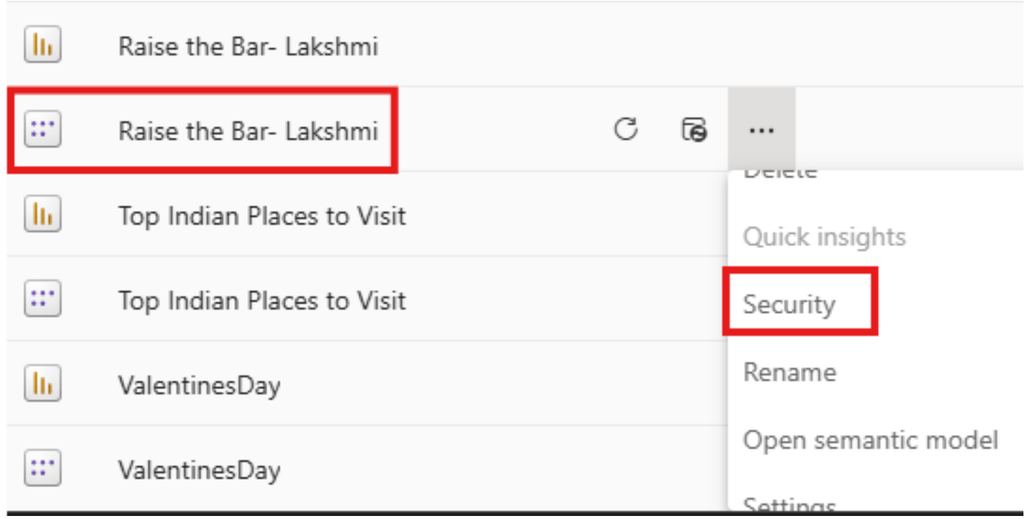


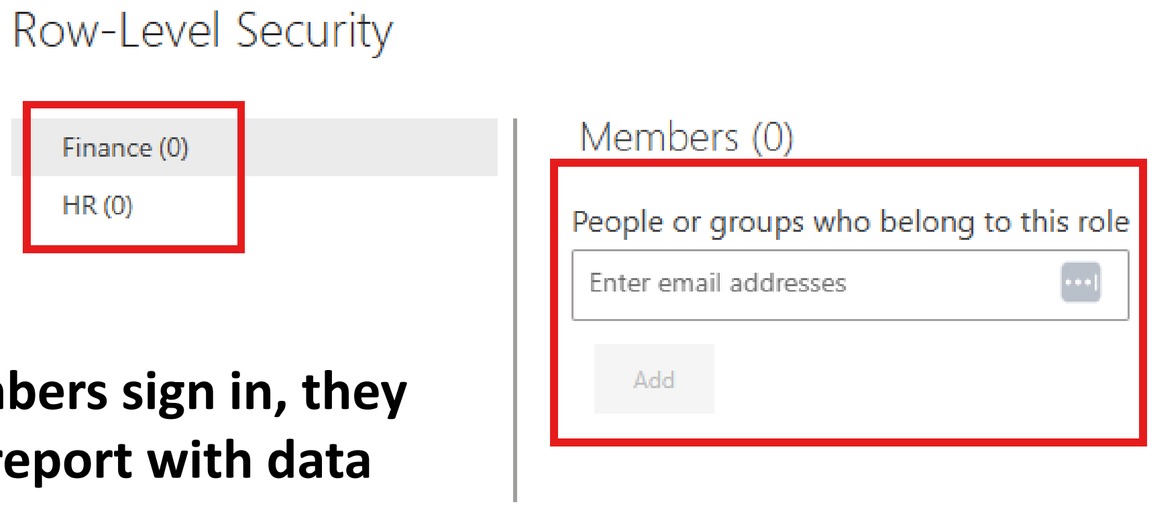
Testing RLS in Power BI Desktop





Add Members to Roles in Power BI Service





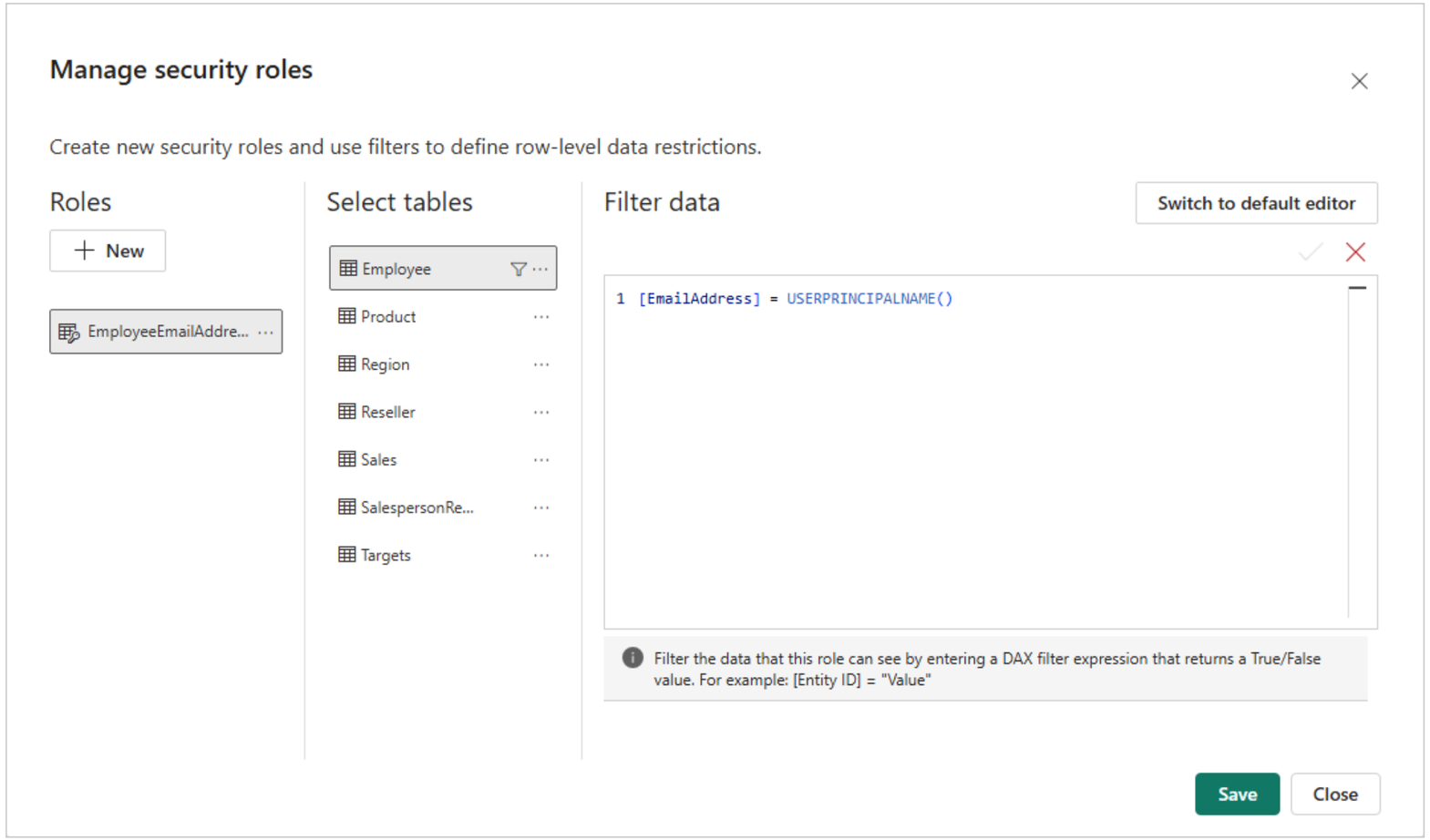
When those members sign in, they will only see the report with data that applies to them.

Dynamic RLS

Defining dynamic rules is simple and effective when a model table stores **username** or **email address values.**

Two major DAX functions:

1. **USERNAME():** (Domain\Username in Desktop) (Email in Service)
2. **USERPRINCIPALNAME**(): (Email in Desktop & Service)

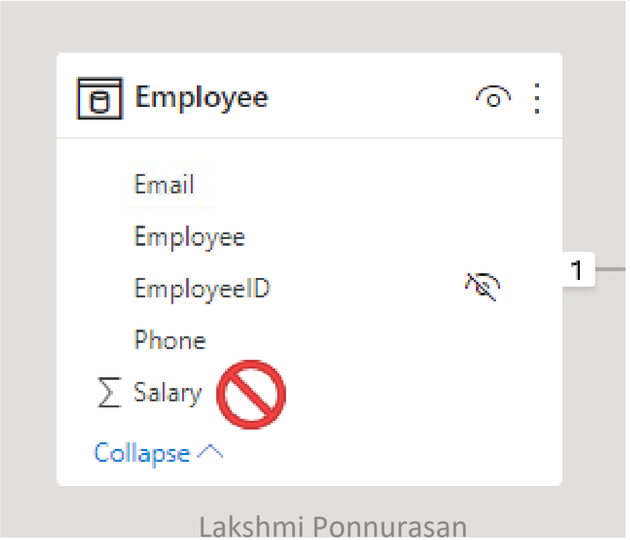


Imagine…

Tailspin Traders has a table named **DimEmployee**. The table includes columns that store employee name, phone, email address, and salary.

While general report consumers can see employee name and contact details, they must not be able to see salary values. Only senior Human Resources staff are permitted to see salary values.

Object Level Security (OLS)



OLS Setup

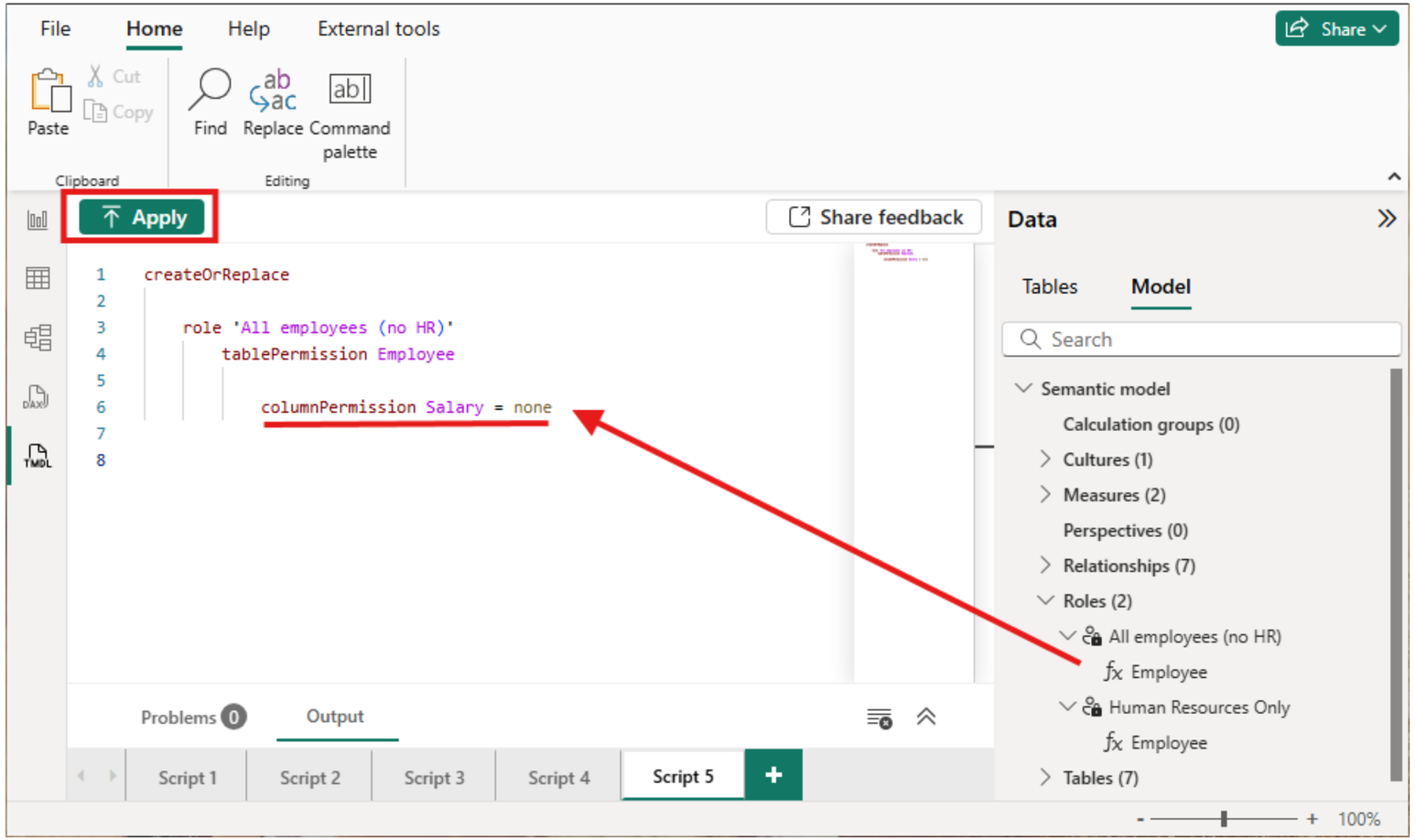
1. To set up OLS, you start by creating roles. (Same as RLS)

2. Next, you need to add OLS rules to the roles.

3. Use TMDL Editor (preview)

   You can set them to **None** or **Read**.
   None- users associated with the role can't access the object.
   Read- users associated with the role can access the object.

   When you're restricting specific columns, ensure the table isn't set to **None**.



Restrictions of OLS

Thank You